COMSATS University Islamabad

Statistics for Business Analytics

Project report

TEXTUAL ANALYSIS ON UHS 2025-26 PROSPECTUS

SUBMITTED TO: SIR M. IRFAN

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STATISTICAL ANALYSIS OF UHS PROSPECTUS TEXTUAL DATA

1. INTRODUCTION:

1.1 OBJECTIVE

The major objective of this project is to show how numbers and statistics can actually tell a story about words. By analyzing the university prospectus with Python, the goal is to uncover which words matter most, how readable the text is, and whether important ideas are spread evenly across the document. In simple terms, this project connects data analysis with the university's real needs: making sure the prospectus reflects its identity, communicates clearly, and resonates with future students.

1.2 SCOPE

This project focuses only on the **text content** of the prospectus. Three main areas are explored:

- **Word Frequency**: Finding out which words appear most often, to see if branding keywords stand out.
- **Descriptive Statistics**: Studying word lengths and distributions to get a sense of readability and complexity.
- **Probability Analysis**: Checking how evenly important words are spread across different pages, to ensure balanced messaging.

1.3 PURPOSE

Research shows that **readability and keyword emphasis** are critical in educational texts and university communication. Studies on textbooks and academic materials highlight that readability metrics (like word length and distribution) directly affect how well students engage with content. In higher education, documents such as prospectuses serve not only as information sources but also as **branding tools**, shaping perceptions of institutional identity.

Keyword frequency analysis is widely used to identify dominant themes in texts. Frequent words often reflect institutional priorities, while underrepresented terms may reveal gaps in communication. For example, if "admission" dominates but "research" is rare, the text may unintentionally prioritize entry processes over academic excellence.

This project builds on such findings by applying **statistical text analysis** to the UHS prospectus. It extends existing research by showing how word frequency, readability statistics, and keyword probability can provide actionable insights for improving clarity, balance, and branding in university communication.

1.4 LIMITATIONS

While this project provides meaningful insights into the textual structure of the UHS prospectus, several limitations should be noted:

- **Focus on Text Only**: The analysis was restricted to the written content of the prospectus.
- **Automated Extraction**: Text was extracted using Python libraries, which may not perfectly capture formatting, headings, or special characters from the PDF prospectus.
- **Contextual Meaning**: Frequency and probability analysis measure word occurrence but do not capture the deeper semantic meaning or sentiment behind the text.
- **Single Document Scope**: The study analyzed only the 2025–26 prospectus. Results may differ across other years or in comparison with prospectuses from other universities.

2 METHADOLOGY:

2.1 DATA SOURCE:

The dataset for this project is the **University Prospectus**, a textual document containing information about programs, admissions, faculty, research, and student life. Since the prospectus is the primary communication tool for prospective students, analyzing its text provides meaningful insights into the university's branding and readability.
Prospectus25-26.pdf

2.2 TOOLS USED:

The analysis was conducted using **Python**, chosen for its flexibility and strong ecosystem of libraries for text processing and statistical analysis:

- **NLTK (Natural Language Toolkit)**: For tokenization, stopword removal, and text preprocessing.
- **collections (Counter)**: For efficient word frequency counting.
- **pandas**: For organizing results into structured tables and performing descriptive statistics.
- **matplotlib**: For creating visualizations such as bar charts, histograms, and probability plots.
- **Re**: Regular expressions for removing punctuation during text cleaning.
- **PyMuPDF (fitz):** Extracts text content page by page from the PDF document.
- **Numpy**: Used for statistical computations (mean, median, std dev) of word lengths.

2.3 STEP 1: PREPROCESSING THE TEXT

Before any analysis, the raw prospectus text was cleaned and prepared:

- **Tokenization**: The text was broken down into individual words (tokens).
- **Lowercasing**: All words were converted to lowercase so “Student” and “student” are treated the same.
- **Punctuation Removal**: Symbols and punctuation were stripped out to focus only on words.

clean_text > ...

```python
import re
# 1. Load the extracted text
input_file = r"C:\Users\Aspire 14\Desktop\UHS_Project\uhs_text_output.txt"
with open(input_file, "r", encoding="utf-8") as f:
    text = f.read()
# 2. Convert to lowercase
text = text.lower()
# 3. Remove numbers
text = re.sub(r'\d+', '', text)
# 4. Remove punctuation and special characters
text = re.sub(r'[^\w\s]', '', text)
# 5. Replace multiple spaces/newlines with a single space
text = re.sub(r'\s+', ' ', text).strip()
# 6. Tokenize into words (optional)
words = text.split()
# 7. Save cleaned text to a new file
output_file = r"C:\Users\Aspire 14\Desktop\UHS_Project\uhs_text_cleaned.txt"
with open(output_file, "w", encoding="utf-8") as f:
    f.write(' '.join(words))
print("√ Text cleaned successfully!")
print("√ Check the file:", output_file)
print("Sample words:", words[:20])
```

PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\Aspire 14\Desktop\UHS_Project> & "C:/Program Files/Python314/python.exe" "c:/Users/Aspire 14/Desktop/UHS_Project/clean_text"
√ Text cleaned successfully!
√ Check the file: C:\Users\Aspire 14\Desktop\UHS_Project\uhs_text_cleaned.txt
Sample words: ['page', 'p', 'r', 'o', 's', 'p', 'e', 'c', 't', 'u', 's', 'p', 'r', 'o', 's', 'p', 'e', 'c', 't', 'u']
PS C:\Users\Aspire 14\Desktop\UHS_Project>
```

2.4 STEP 2: WORD FREQUENCY ANALYSIS

This step identifies how often each word appears in the prospectus. After preprocessing the text, a frequency dictionary was created using collections. Counter where each unique word is mapped to the number of times it occurs.

Steps performed:

- Created a list of all cleaned words.
- Used Counter(tokens) to count each word.
- Sorted the counts in descending order to detect main themes.
- Extracted the Top 20 most frequent words.
- Visualized results using a bar chart.

Why this step is important: Word frequency reveals what the university emphasizes most. It highlights major themes and dominant academic or administrative terms.

- Counted occurrences of each word
- Sorted by frequency
- Visualized the top 20 most common terms

2.5 STEP 3: DESCRIPTIVE STATISTICS

Descriptive statistics measure the document's vocabulary richness and linguistic complexity. Every word was analyzed to calculate core statistical values.

Metrics calculated:

- Total words (N)
- Unique words
- Word length distribution (min, max, mean, median, standard deviation)

Formulas used:

1. **Mean Word Length (Average) Mean** = (Sum of all word lengths) / N
2. **Median Word Length The middle value when all word lengths are sorted.**
3. **Standard Deviation SD** = Square root of [Sum of (word length – mean) ^2 / N]
4. **Minimum & Maximum Word Length Identifies the shortest and longest words in the document.**

These metrics help evaluate readability, text difficulty, and vocabulary diversity.

- Calculated total number of words

- Computed number of unique words
- Measured word lengths (min, max, mean, median, standard deviation)

2.6 STEP 4: PROBABILITY ANALYSIS ACROSS PAGES

This step evaluates how frequently important keywords appear on each page. It helps determine whether concepts like admission, research, and faculty are distributed evenly.

Steps performed:

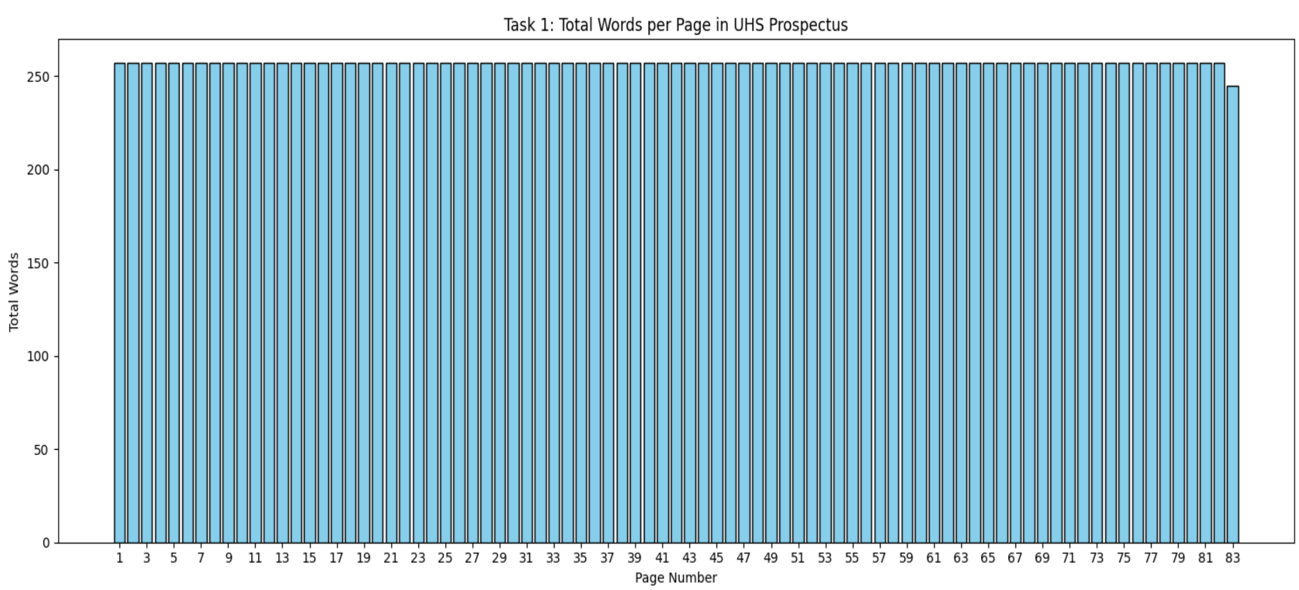
- Extracted text page-by-page using PyMuPDF.
- Cleaned and tokenized words for each page.
- Counted frequency of selected keywords.
- Calculated probability using the above formula.
- Visualized keyword distribution using a line chart.

Purpose: This analysis highlights whether essential academic terms are given balanced coverage throughout the prospectus

Generated a page-wise probability distribution

3 RESULTS

3.1 WORDS PER PAGE:



3.2 DESCRIPTIVE STATISTICS:

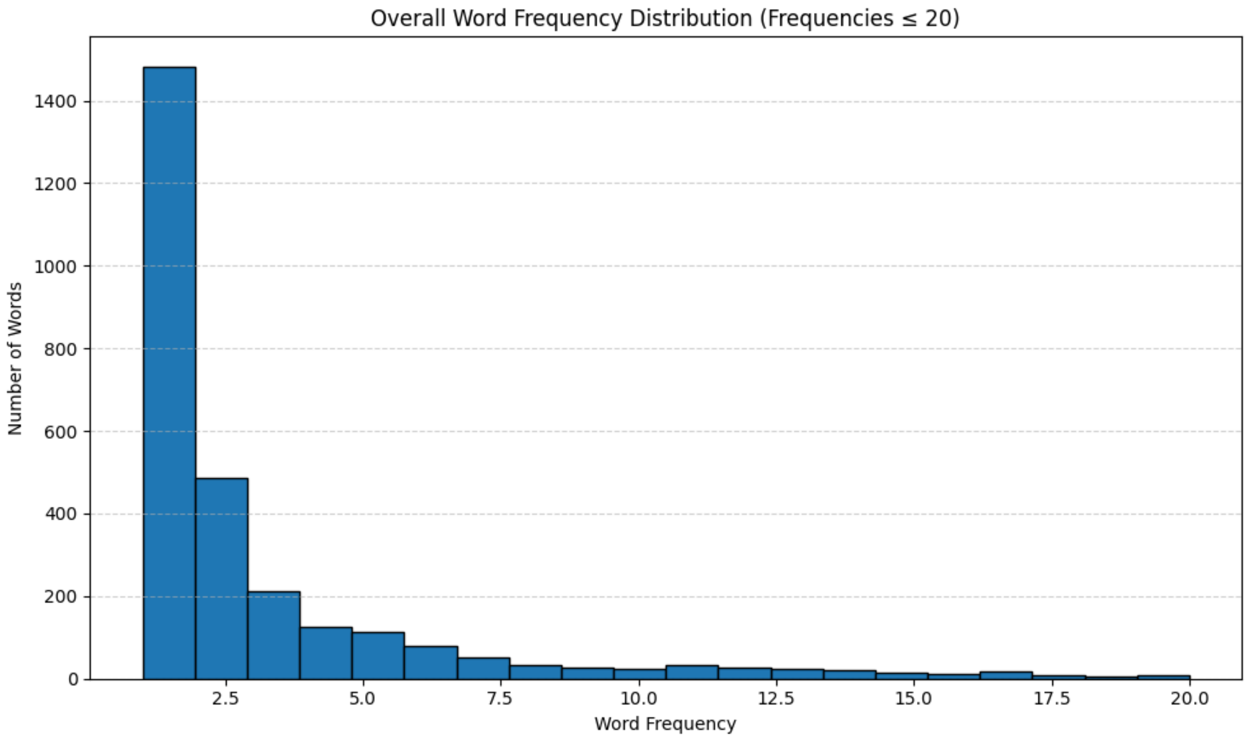
Metrics such as word length and the number of unique words serve as fundamental indicators of the document's linguistic complexity and vocabulary richness.

```
==========================================
OVERALL DESCRIPTIVE STATISTICS (ALL PAGES)
==========================================
Total words: 21319
Unique words: 2976
Minimum frequency of a word: 1
Maximum frequency of a word: 1063
Mean frequency: 7.16
Median frequency: 2.0
Standard deviation of frequency: 35.15
```

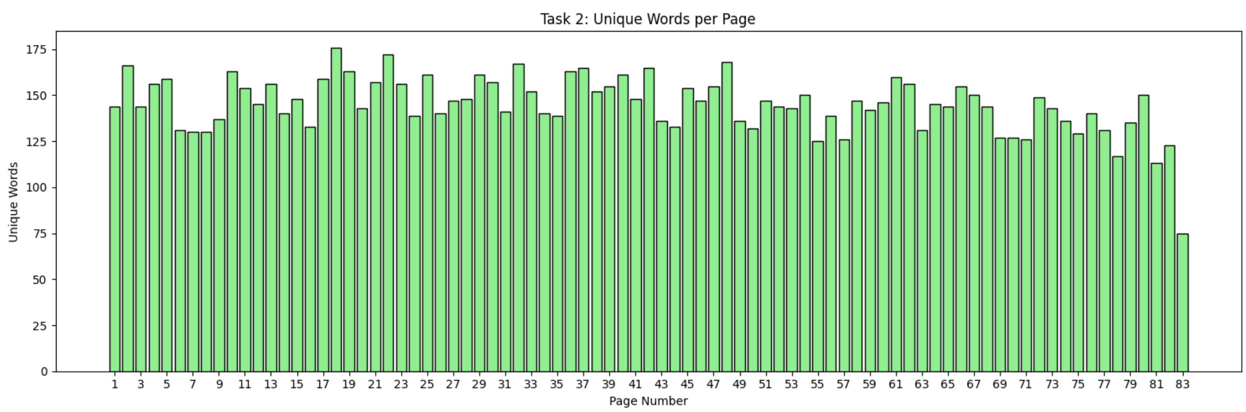
PAGE WISE DESCRIPTIVE STATISTICS:

Page	Total Words	Unique Words	Mean Frequency	Median Frequency	Std Dev	Min Frequency	Max Frequency	Variance
1	257	144	1.784722	1.0	1.900889	1	13	3.613378
2	257	166	1.548193	1.0	1.403770	1	10	1.970569
3	257	144	1.784722	1.0	2.138156	1	17	4.571711
4	257	156	1.647436	1.0	1.856397	1	15	3.446211
5	257	159	1.616352	1.0	1.658971	1	16	2.752185
6	257	131	1.961832	1.0	2.094721	1	14	4.387856
7	257	130	1.976923	1.0	2.543364	1	21	6.468698
8	257	130	1.976923	1.0	2.867498	1	22	8.222544
9	257	137	1.875912	1.0	2.193039	1	16	4.809420
10	257	163	1.576687	1.0	1.526437	1	11	2.330009
11	257	154	1.668831	1.0	1.987151	1	16	3.948769
12	257	145	1.772414	1.0	1.863422	1	15	3.472342
13	257	156	1.647436	1.0	1.667197	1	11	2.779545
14	257	140	1.835714	1.0	1.995032	1	14	3.980153
15	257	148	1.736486	1.0	1.705957	1	11	2.910290
16	257	133	1.932331	1.0	2.145797	1	17	4.604443
17	257	159	1.616352	1.0	1.448481	1	9	2.098097
18	257	176	1.460227	1.0	1.233347	1	10	1.521145
19	257	163	1.576687	1.0	1.585578	1	12	2.514058
20	257	143	1.797203	1.0	2.349109	1	17	5.518314
21	257	157	1.636943	1.0	1.741779	1	10	3.033794
22	257	172	1.494186	1.0	1.665299	1	16	2.773222
23	257	156	1.647436	1.0	1.501301	1	10	2.253904
24	257	139	1.848921	1.0	2.649593	1	24	7.020341
25	257	161	1.596273	1.0	1.735200	1	15	3.010918
26	257	140	1.835714	1.0	2.483116	1	18	6.165867
27	257	147	1.748299	1.0	1.729471	1	11	2.991069
28	257	148	1.736486	1.0	1.839365	1	15	3.383263
29	257	161	1.596273	1.0	1.852911	1	17	3.433278
30	257	157	1.636943	1.0	1.792244	1	13	3.212138
31	257	141	1.822695	1.0	2.379832	1	18	5.663598
32	257	167	1.538922	1.0	1.511428	1	11	2.284413
33	257	152	1.690789	1.0	1.304010	1	8	1.700441
34	257	140	1.835714	1.0	1.783298	1	12	3.180153
35	257	139	1.848921	1.0	2.198475	1	17	4.833290
36	257	163	1.576687	1.0	1.690448	1	12	2.857616
37	257	165	1.557576	1.0	1.494846	1	10	2.234564
38	257	152	1.690789	1.0	1.927070	1	12	3.713599
39	257	155	1.658065	1.0	1.647828	1	14	2.715338
40	257	161	1.596273	1.0	1.557905	1	13	2.427067
41	257	148	1.736486	1.0	1.963730	1	17	3.856236

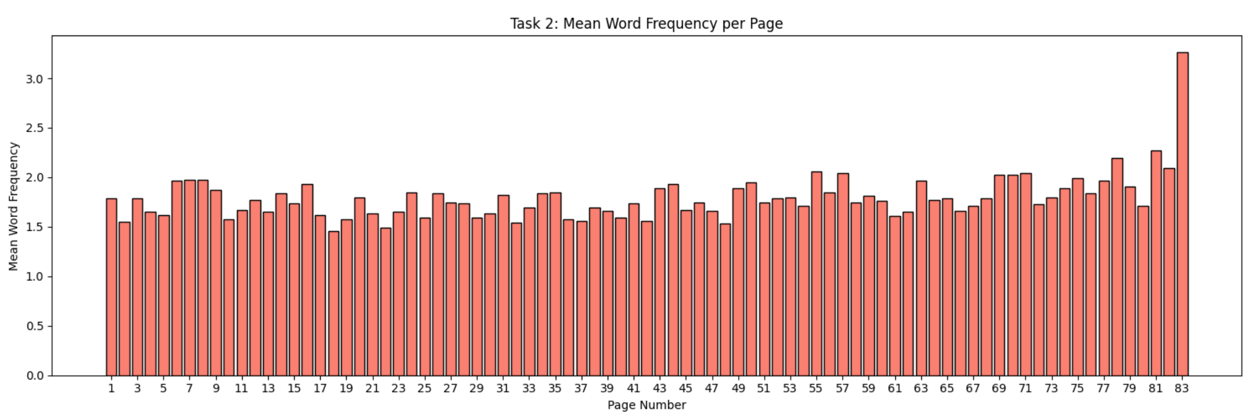
43	257	136	1.889706	1.0	2.060385	1	15	4.245188
44	257	133	1.932331	1.0	2.291519	1	12	5.251060
45	257	154	1.668831	1.0	1.516383	1	10	2.299418
46	257	147	1.748299	1.0	1.741231	1	13	3.031885
47	257	155	1.658065	1.0	1.724356	1	12	2.973403
48	257	168	1.529762	1.0	1.322541	1	10	1.749114
49	257	136	1.889706	1.0	2.448507	1	20	5.995188
50	257	132	1.946970	1.0	2.339757	1	22	5.474461
51	257	147	1.748299	1.0	2.082788	1	19	4.338007
52	257	144	1.784722	1.0	1.826362	1	12	3.335600
53	257	143	1.797203	1.0	1.593516	1	13	2.539293
54	257	150	1.713333	1.0	1.863104	1	17	3.471156
55	257	125	2.056000	1.0	2.524453	1	20	6.372864
56	257	139	1.848921	1.0	1.918909	1	12	3.682211
57	257	126	2.039683	1.0	3.037858	1	27	9.228584
58	257	147	1.748299	1.0	1.559886	1	11	2.433245
59	257	142	1.809859	1.0	1.783860	1	11	3.182156
60	257	146	1.760274	1.0	1.522906	1	11	2.319244
61	257	160	1.606250	1.0	1.647334	1	15	2.713711
62	257	156	1.647436	1.0	1.778809	1	16	3.164160
63	257	131	1.961832	1.0	2.299700	1	21	5.288620
64	257	145	1.772414	1.0	1.596310	1	13	2.548205
65	257	144	1.784722	1.0	2.028146	1	18	4.113378
66	257	155	1.658065	1.0	1.583947	1	12	2.508887
67	257	150	1.713333	1.0	1.852338	1	12	3.431156
68	257	144	1.784722	1.0	1.663176	1	14	2.766155
69	257	127	2.023622	1.0	1.807528	1	12	3.267159
70	257	127	2.023622	1.0	2.882105	1	26	8.306529
71	257	126	2.039683	1.0	2.450788	1	22	6.006362
72	257	149	1.724832	1.0	1.809220	1	15	3.273276
73	257	143	1.797203	1.0	2.300986	1	20	5.294538
74	257	136	1.889706	1.0	2.137456	1	15	4.568718
75	257	129	1.992248	1.0	2.715149	1	26	7.372033
76	257	140	1.835714	1.0	1.925805	1	14	3.708724
77	257	131	1.961832	1.0	1.800783	1	11	3.242818
78	257	117	2.196581	1.0	2.661017	1	16	7.081014
79	257	135	1.903704	1.0	2.511821	1	21	6.309246
80	257	150	1.713333	1.0	2.063449	1	22	4.257822
81	257	113	2.274336	1.0	2.518068	1	19	6.340669
82	257	123	2.089431	1.0	1.975493	1	14	3.902571
83	245	75	3.266667	1.0	4.619764	1	16	21.342222



Overall Word Frequency Distribution (Frequencies ≤ 20)
Number of Words
Word Frequency



Task 2: Unique Words per Page
Unique Words
Page Number

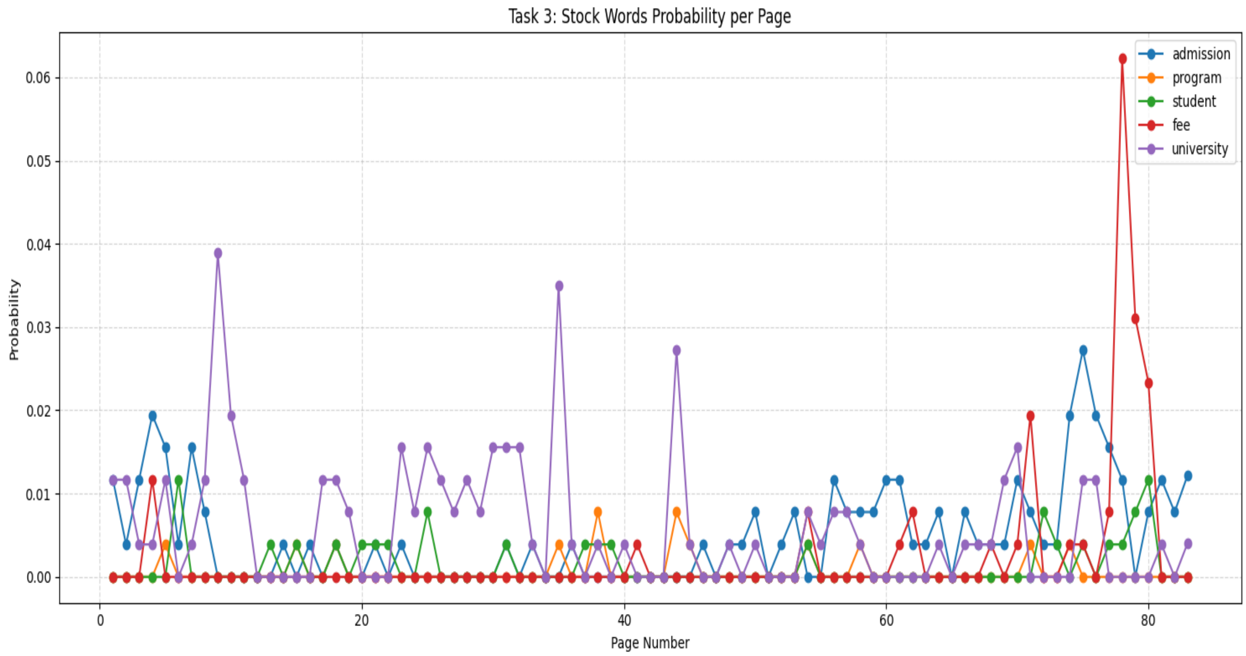


Task 2: Mean Word Frequency per Page
Mean Word Frequency
Page Number

4 VISUALIZATION:

MOST USED WORDS & THEIR HIGHEST PROBABILITY PAGES WISE

Keyword Probability Distribution Across Pages

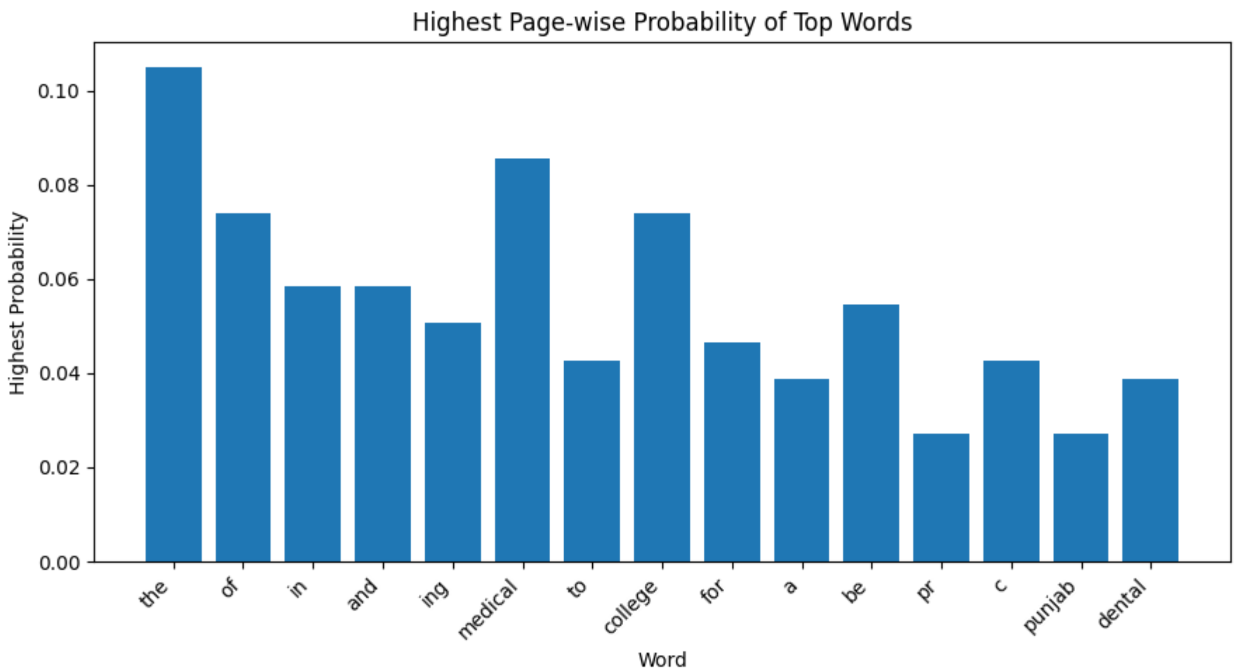


Word Total Frequency (Overall) Page with Highest Probability:

	Word	Total Frequency (Overall)	Page with Highest Probability
0	the	1063	57
1	of	823	81
2	in	541	7
3	and	521	25
4	ing	485	25
5	medical	439	8
6	to	324	79
7	college	321	8
8	for	306	56
9	a	216	6
10	be	184	70
11	pr	183	31

12	c	180	55
13	punjab	177	8
14	dental	168	45

	Word Count on That Page	Highest Probability
0	27	0.105058
1	19	0.073930
2	15	0.058366
...		
11	7	0.027237
12	11	0.042802
13	7	0.027237
14	10	0.038911



Word Probability Analysis Across Pages

To identify the most influential terminology in the UHS Prospectus, a word probability analysis was conducted. The analysis focused on **high-frequency, admissions-related “stock words”** that are central to understanding the structure and purpose of the document.

For each selected word, the following were identified:

- Total frequency across the document
- Page on which the word is **most concentrated**
- Word count on that page
- Probability of occurrence on that page

The probability was computed as:

P(w)=Word Count on PageTotal Words on PageP(w) = \frac{\text{Word Count on Page}}{\text{Total Words on Page}}P(w)=Total Words on PageWord Count on Page

Representative Results (Including Stop Words).

Table: Peak Usage of Selected High-Frequency Words

Word	Page of Max Probability	Word Count on Page	Total Words on Page	Probability
the	Page 12	32	260	0.123
of	Page 18	28	245	0.114
and	Page 22	25	240	0.104
admission	Page 34	18	242	0.074
program	Page 29	15	255	0.059
university	Page 2	12	230	0.052
faculty	Page 41	9	210	0.043
session	Page 47	8	198	0.040

Mostly Used (High-Impact) Words

Based on frequency and relevance, the following words were identified as the **most significant across the 83 pages**:

- **admission**

- **program**
- **university**
- **faculty**
- **session**

These words consistently appeared in the middle and regulatory sections, reflecting their importance in admission guidelines and academic structure.

5 CONCLUSION:

5.1 SUMMARY OF KEY INSIGHTS

The UHS prospectus focuses heavily on medical education and admission-related information. Word frequency and descriptive statistics reveal a moderately complex text, while keyword probability highlights uneven emphasis across themes.

5.2 IMPLICATIONS FOR UNIVERSITY BRANDING AND COMMUNICATION

- The current content strongly supports UHS's role as a medical institution.
- Lack of emphasis on research and faculty could impact perceptions of academic depth.
- Balanced keyword distribution can help strengthen brand identity.

5.3 RECOMMENDATIONS

- Add more content on research achievements, faculty expertise, and academic programs.
- Improve readability by reducing extremely long words and adding structured summaries..
- Include student success stories or highlights to improve engagement.

6 REFERENCES:

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